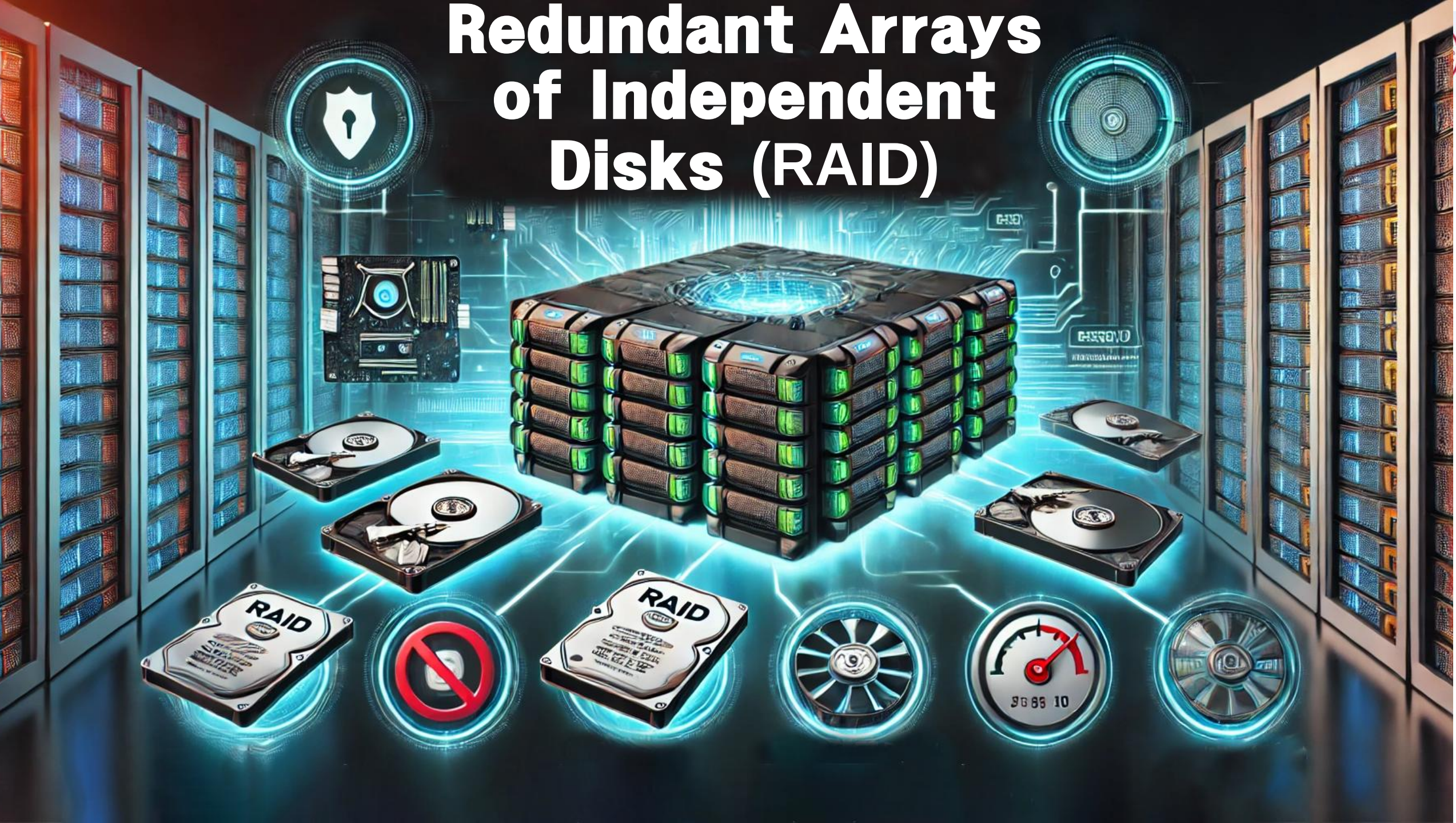
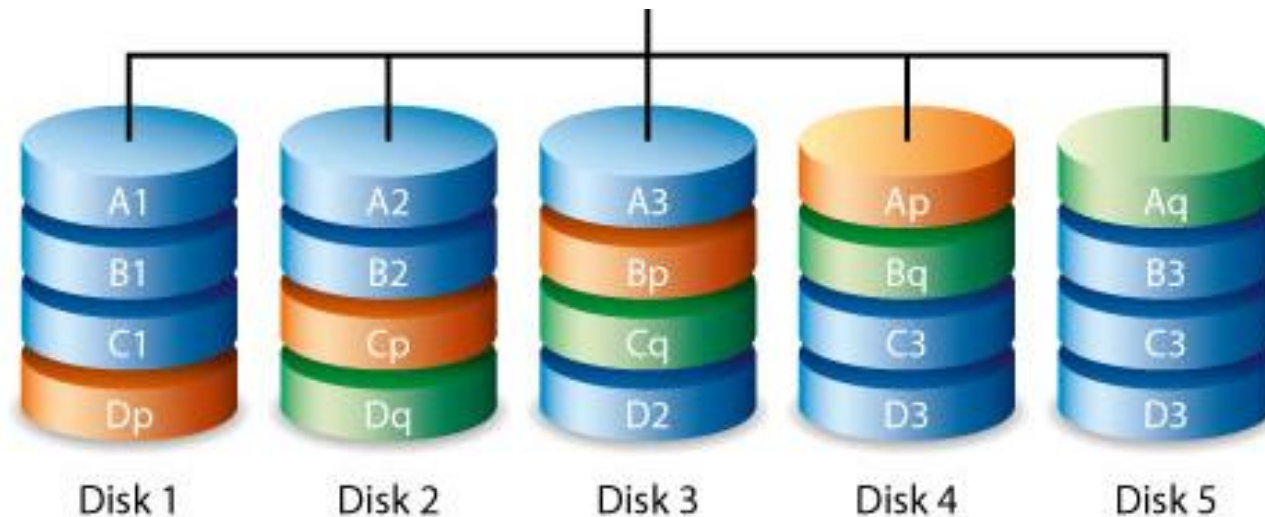


Redundant Arrays of Independent Disks (RAID)



- **Redundant Arrays of Independent Disks (RAID)**
 - Disk organization techniques
 - Manage a large numbers of disks,
 - Providing a view of a single disk of
 - **High capacity** and **high speed** by using **multiple disks** in **parallel**,
 - **High reliability** by storing data redundantly



RAID



$$\text{RAID} = \begin{array}{|c|} \hline 32 \text{ GB} \\ \hline \end{array} + \begin{array}{|c|} \hline 32 \text{ GB} \\ \hline \end{array}$$



RAID Levels

- Schemes to provide redundancy at lower cost by using disk striping combined with parity bits.
- Performance metrics:
 - **Cost,**
 - **Performance**
 - **Reliability.**
- RAID levels:
 - RAID Level 0
 - RAID Level 1
 - RAID Level 2
 - RAID Level 3
 - RAID Level 4
 - RAID Level 5
 - RAID Level 6

RAID Levels

- **RAID Level 0:**

- **Block striping;**

- Segmenting logically sequential data, such as a file,
 - Consecutive segments are stored on different physical storage devices.

- Used in high-performance applications where data loss is not critical.

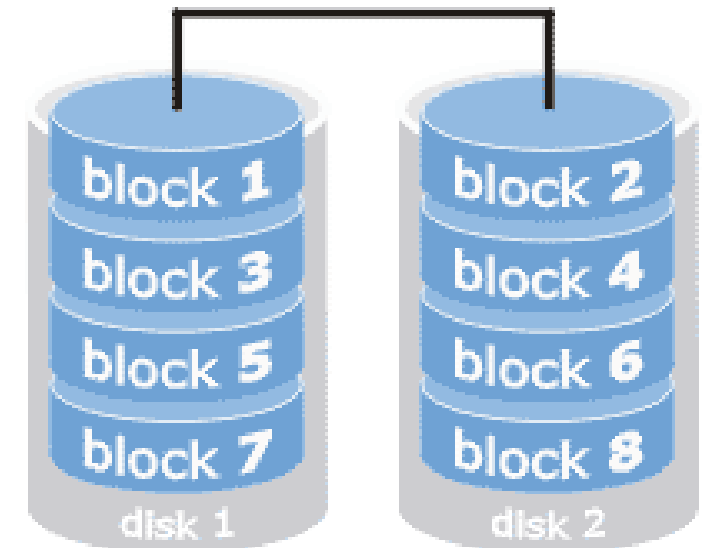
- No Mirror, No parity

- Minimum 2 disks

- Excellent performance

- No redundancy

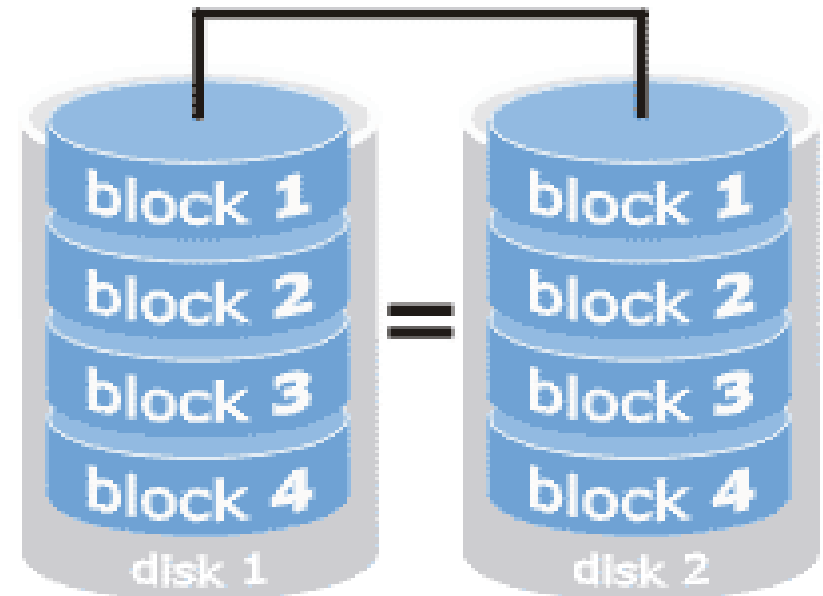
- Don't use it for any critical system



RAID Levels

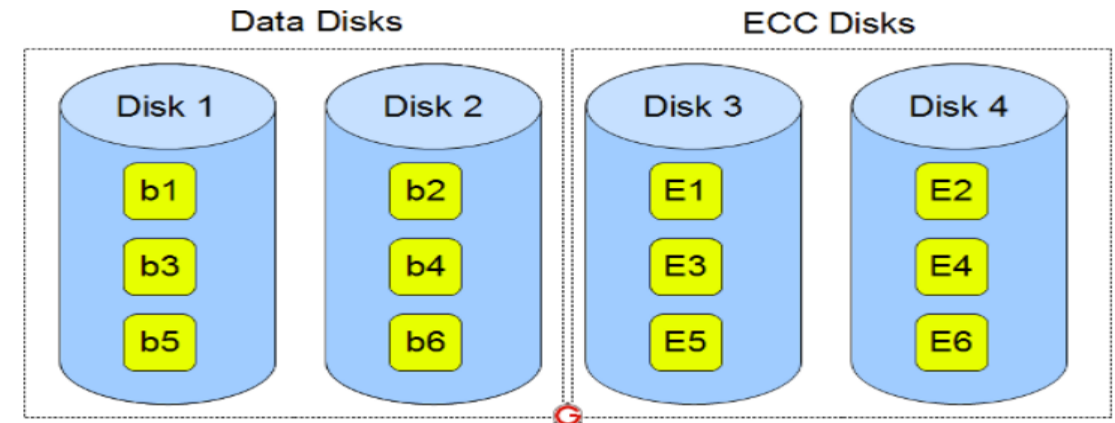
- **RAID Level 1**

- **Mirrored** disks with block striping
- Offers best write performance.
- Popular for applications such as storing log files in a database system.
- No stripe
- No parity
- Minimum 2 disks
- Good performance
- Excellent redundancy



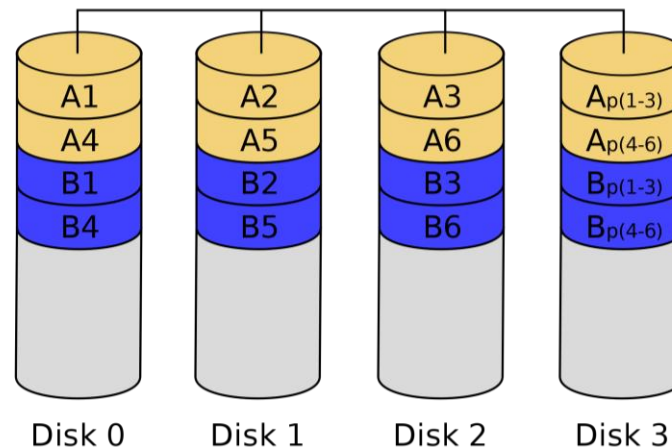
RAID Levels

- **RAID Level 2:**
 - **Memory-Style Error-Correcting-Codes (ECC)** with **bit striping**.
 - b1, b2, b3: to write the data
 - E1, E2, E3: to write the error correction codes
 - **Reading:** the reader reads both data and correction codes.
 - **Data recovery:** codes are applied to correct data.



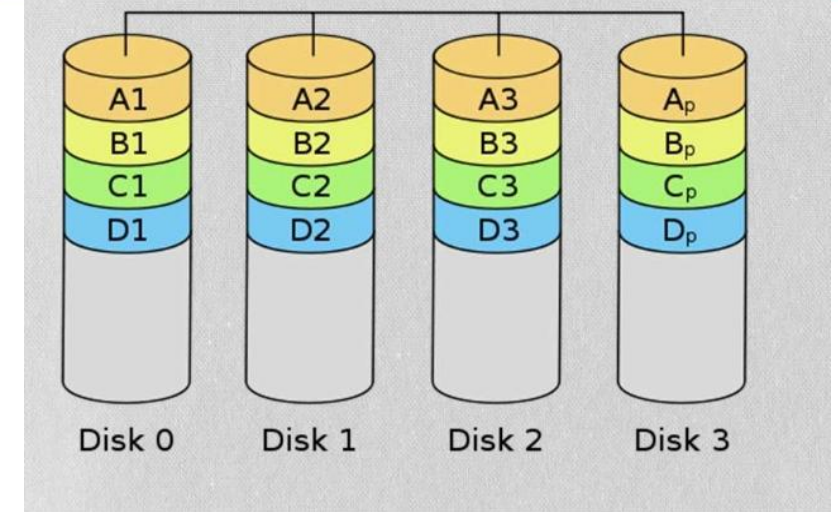
RAID Levels

- **RAID Level 3:**
 - **Bit-Interleaved Parity**
 - A single **parity bit** for error correction
 - **Writing:** compute and write the corresponding parity bits to a parity bit disk
 - Data recovery: compute XOR of bits from other disks (including parity bit disk)



RAID Levels

- **RAID Level 4:**
 - **Block-Interleaved Parity**
 - **Block-level striping,**
 - Keeps a parity block on a separate disk for corresponding blocks from N other disks.
 - **Writing:** compute and write the corresponding block of parity bits to parity disk
 - **Data recovery:** compute XOR of bits from corresponding blocks (including parity block) from other disks.
- RAID 4 is **not** commonly used.



RAID Levels

- RAID Level 5



RAID 5: block-interleaved distributed parity

- Block-Interleaved Distributed Parity

- Partitions data and parity among all $N + 1$ disks.

- e.g., with 5 disks:

P0	0	1	2	3
4	P1	5	6	7
8	9	P2	10	11
12	13	14	P3	15
16	17	18	19	P4

- Parity block for n^{th} set of blocks is stored on disk $(n \bmod 5) + 1$, with the data blocks stored on the other 4 disks.

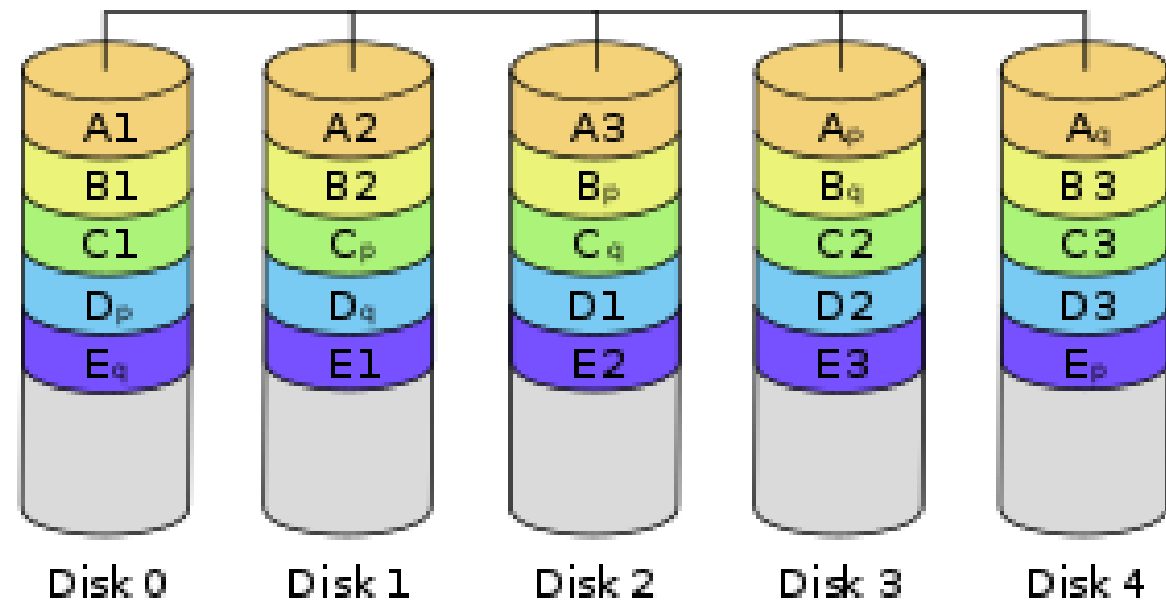
- Minimum 3 disks
- Good performance
- Good redundancy
- Best cost effective

RAID Levels

- **RAID Level 6**

- **P+Q Redundancy scheme**

- Stores extra redundant information to guard against multiple disk failures.
 - Better reliability than Level 5
 - Higher cost;
 - Not used as widely.



RAID Selection

- **RAID 0:** if data safety is not important
 - e.g. data can be recovered quickly from other sources
- **Level 2** and **4:** never used
- **Level 3:** not used anymore since
- **Level 6** is rarely used
- **Levels 1** and **5** offer adequate safety for most applications

Implementation

1. Software RAID:

- RAID implementations done entirely in software, with no special hardware support.

2. Hardware RAID:

- RAID implementations with special hardware
- Use **non-volatile RAM** to record writes that are being executed
- Beware: power failure during write can result in corrupted disk

Hardware Issues

- **Latent failures:** data successfully written earlier gets damaged
 - Can result in data loss even if only one disk fails
- **Data scrubbing:**
 - Continually scan for latent failures, and recover from copy/parity
- **Hot swapping:** replacement of disk while system is running, without power down
 - Supported by some hardware RAID systems,
 - Reduces time to recovery, and improves availability greatly
- Many systems maintain **spare disks** which are kept online, and used as replacements for failed disks immediately on detection of failure
 - Reduces time to recovery greatly
- Many hardware RAID systems ensure that a single point of failure will not stop the functioning of the system by using
 - Redundant power supplies with battery backup
 - Multiple controllers and multiple interconnections to guard against controller/interconnection failures



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